

Gauge R&R Study Process Flow

Steps for conducting a complete MSA study

1. Plan Study

Define measurement, tolerance, study type (variable/attribute)

2. Select Parts

Select 10 parts spanning process range

3. Identify Appraisers

Select 2–3 operators who normally use gauge

4. Calibrate Gauge

Verify gauge is calibrated and in good condition

5. Randomize Order

Randomize measurement order to prevent bias

6. Conduct Measurements

Each operator measures each part 2–3 times (blind)

7. Analyze Data

Calculate EV, AV, GRR, %GRR, ndc

8. Interpret Results

Compare to acceptance criteria

9. Take Action

If unacceptable, identify root cause and improve

10. Document

Record results in quality system

